

Problem Set 1 - Suggested Solutions

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1 Sum Stats and Institutional Details

1.1 Do some brief diligence on the products and industry. What do you anticipate may be some important determinants of demand, substitution, and pricing? (5 points)

5 points for reasonable answer

Answers may vary. Students may include information on conditions treated by the medications, the prevalence of these conditions, or relevant (perceived or otherwise) differences between the products. Responses may also include general information on popularity or use trends for the products.

1.2 Complete the table above by adding columns for the mean: market share, unit price, price/100 tablets, and unit wholesale price. Interpret any notable patterns you see in the summary statistics. (10 points)

5 points for reasonable code

5 points for accurate table

```
[1]: import numpy as np
import pandas as pd
from linearmodels.iv import IV2SLS
import statsmodels.api as sm
import statsmodels.formula.api as smf
otc = pd.read_csv('OTC_Sales.csv')
```

```
[2]: # calculate total sales for each store-week pair
product_numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
otc['total_sales'] = otc[[f'sales_{number}' for number in product_numbers]].
    ↪sum(axis = 1)
```

```
[3]: # reshape to make it easier to work with
otc = pd.wide_to_long(otc, ['sales', 'price', 'cost', 'prom'],
                      i = ['store', 'week'], j = 'product', sep = '_').
    ↪reset_index()
summary = otc.groupby('product').agg(
    price = ('price', 'mean'),
    total_sales = ('sales', 'sum'),
```

```

    wholesale_price = ('cost', 'mean')
).reset_index()

# add given information so that table is easy to read
brand_names = {
    1: 'Tylenol',
    2: 'Tylenol',
    3: 'Tylenol',
    4: 'Advil',
    5: 'Advil',
    6: 'Advil',
    7: 'Bayer',
    8: 'Bayer',
    9: 'Bayer',
    10: 'Store Brand',
    11: 'Store Brand'}

size = {
    1: 25,
    2: 50,
    3: 100,
    4: 25,
    5: 50,
    6: 100,
    7: 25,
    8: 50,
    9: 100,
    10: 50,
    11: 100
}

summary['brand'] = summary['product'].map(brand_names)
summary['size'] = summary['product'].map(size)

# calculate new variables
summary['market_share'] = summary['total_sales'] / (summary['total_sales'].
↪sum())
summary['price_per_100'] = (summary['price'] / summary['size']) * 100

# reorder columns and round for clarity
table_1 = summary[['product', 'brand', 'size', 'market_share',
                    'price', 'price_per_100', 'wholesale_price']].round(2)
table_1

```

```

[3]:
   product  brand  size  market_share  price  price_per_100  \
0         1  Tylenol    25          0.14    3.43          13.71
1         2  Tylenol    50          0.18    4.95           9.89
2         3  Tylenol   100          0.12    7.03           7.03
3         4   Advil    25          0.12    2.97          11.88

```

4	5	Advil	50	0.08	5.15	10.29
5	6	Advil	100	0.04	8.16	8.16
6	7	Bayer	25	0.04	2.67	10.70
7	8	Bayer	50	0.03	3.62	7.24
8	9	Bayer	100	0.08	3.97	3.97
9	10	Store Brand	50	0.09	1.94	3.87
10	11	Store Brand	100	0.08	4.43	4.43

	wholesale_price
0	2.19
1	3.68
2	5.77
3	2.03
4	3.63
5	6.10
6	1.85
7	2.44
8	3.71
9	0.91
10	1.91

2 Logit Demand Estimation

Consider the utility function for product j in store-week t for consumer i :

$$u_{ijt} = -p_{jt} + X_{jt} + \xi_{jt} + \epsilon_{ijt} \quad (1)$$

where p_{jt} is price, X_{jt} are other observed product characteristics, ξ_{jt} are unobserved product characteristics, and ϵ_{ijt} is an i.i.d. EV1 logit consumer-product-market unobservable.

Estimate this model:

2.1 Using OLS with price, promotion, and an indicator for whether the product is a “store brand” as product characteristics. (10 points)

5 points for correct independent variables

2 points for correct dependent variables

3 points for reasonable results

```
[4]: # need shares for each observation
otc['share'] = otc['sales'] / otc['count']
otc['outside_share'] = 1 - (otc['total_sales'] / otc['count'])

# indicator for store brand
otc['store_brand'] = np.where(otc['product'] > 9, 1, 0)
otc['size'] = otc['product'].map(size)
```

```
[5]: # create dependent variable: ln(share - outside_share)
otc['ln_share'] = np.log(otc['share']) - np.log(otc['outside_share'])
```

```
[6]: model = smf.ols('ln_share ~ price + prom + store_brand', data = otc)
logit_modela = model.fit()
logit_modela.summary()
```

[6]:

Dep. Variable:	ln_share	R-squared:	0.026
Model:	OLS	Adj. R-squared:	0.026
Method:	Least Squares	F-statistic:	335.2
Date:	Sun, 02 Mar 2025	Prob (F-statistic):	8.60e-215
Time:	17:38:50	Log-Likelihood:	-48633.
No. Observations:	37290	AIC:	9.727e+04
Df Residuals:	37286	BIC:	9.731e+04
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-7.6453	0.014	-552.117	0.000	-7.672	-7.618
price	-0.0662	0.003	-24.543	0.000	-0.071	-0.061
prom	0.1990	0.016	12.316	0.000	0.167	0.231
store_brand	-0.2672	0.013	-21.180	0.000	-0.292	-0.242

Omnibus:	2157.310	Durbin-Watson:	1.362
Prob(Omnibus):	0.000	Jarque-Bera (JB):	2189.616
Skew:	-0.552	Prob(JB):	0.00
Kurtosis:	2.563	Cond. No.	18.2

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

2.2 Using OLS with price and promotion as product characteristics and product fixed effects (where a “product” is a brand-size combination). (10 points)

5 points for correct independent variables

2 points for correct dependent variable

3 points for reasonable results

```
[7]: model = smf.ols('ln_share ~ price + prom + C(product)', data = otc)
logit_modelb = model.fit()
logit_modelb.summary()
```

[7]:

Dep. Variable:	ln_share	R-squared:	0.457
Model:	OLS	Adj. R-squared:	0.457
Method:	Least Squares	F-statistic:	2614.
Date:	Sun, 02 Mar 2025	Prob (F-statistic):	0.00
Time:	17:38:50	Log-Likelihood:	-37744.
No. Observations:	37290	AIC:	7.551e+04
Df Residuals:	37277	BIC:	7.562e+04
Df Model:	12		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-6.0693	0.037	-164.587	0.000	-6.142	-5.997
C(product)[T.2]	0.6879	0.023	30.566	0.000	0.644	0.732
C(product)[T.3]	0.9188	0.040	22.778	0.000	0.840	0.998
C(product)[T.4]	-0.4312	0.017	-25.661	0.000	-0.464	-0.398
C(product)[T.5]	-0.1960	0.024	-8.178	0.000	-0.243	-0.149
C(product)[T.6]	0.0045	0.051	0.088	0.930	-0.096	0.105
C(product)[T.7]	-1.6597	0.018	-93.199	0.000	-1.695	-1.625
C(product)[T.8]	-1.5844	0.016	-96.398	0.000	-1.617	-1.552
C(product)[T.9]	-0.5523	0.018	-31.542	0.000	-0.587	-0.518
C(product)[T.10]	-1.2360	0.022	-56.168	0.000	-1.279	-1.193
C(product)[T.11]	-0.7387	0.019	-38.296	0.000	-0.777	-0.701
price	-0.3404	0.010	-33.375	0.000	-0.360	-0.320
prom	0.3220	0.013	25.508	0.000	0.297	0.347

Omnibus:	2680.344	Durbin-Watson:	1.901
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3703.376
Skew:	-0.621	Prob(JB):	0.00
Kurtosis:	3.916	Cond. No.	112.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

2.3 Estimate the models of (a) and (b) using the Hausman instrument (average price in other markets). (15 points)

5 points for correct calculation of Hausman instrument

6 points for correct 2SLS specification

4 points for reasonable results

```
[8]: def get_instruments(df):
    # compute average prices per store, week, and product
    avg_prices = df.groupby(['store', 'week', 'product'])['price'].mean().
    ↪reset_index()

    # merge to bring average prices into the original dataframe
    df = df.merge(avg_prices, on=['store', 'week', 'product'], suffixes=('_',
    ↪'_avg'))
```

```

    # compute instrument: average price across stores excluding the current_
↪store
    df['instrument'] = df.groupby(['week', 'product'])['price_avg'].
↪transform(lambda x: (x.sum() - x) / (x.count() - 1))

    return df.drop(columns=['price_avg'])

otc = get_instruments(otc)

```

```

[9]: # model a using IV
formula = ('ln_share ~ 1 + prom + store_brand + [price ~ instrument]')
logit_modela_iv = IV2SLS.from_formula(formula, otc).fit()
logit_modela_iv.summary

```

[9]:

Dep. Variable:	ln_share	R-squared:	0.0263
Estimator:	IV-2SLS	Adj. R-squared:	0.0262
No. Observations:	37290	F-statistic:	874.64
Date:	Sun, Mar 02 2025	P-value (F-stat)	0.0000
Time:	17:38:50	Distribution:	chi2(3)
Cov. Estimator:	robust		

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
Intercept	-7.6453	0.0143	-533.36	0.0000	-7.6734	-7.6172
prom	0.1990	0.0158	12.585	0.0000	0.1680	0.2300
store_brand	-0.2672	0.0136	-19.635	0.0000	-0.2938	-0.2405
price	-0.0662	0.0028	-23.364	0.0000	-0.0718	-0.0607

Endogenous: price

Instruments: instrument

Robust Covariance (Heteroskedastic)

Debiased: False

```

[10]: # model b using IV
otc_p = pd.get_dummies(otc, columns = ['product'])
fixed_effects = " + ".join([col for col in otc_p.columns if col.
↪startswith("product_")])

formula = f'ln_share ~ prom + {fixed_effects} + [price ~ instrument]'
logit_modelb_iv = IV2SLS.from_formula(formula, otc_p).fit()
logit_modelb_iv.summary

```

[10]:

Dep. Variable:	ln_share	R-squared:	0.4520
Estimator:	IV-2SLS	Adj. R-squared:	0.4519
No. Observations:	37290	F-statistic:	6.581e+06
Date:	Sun, Mar 02 2025	P-value (F-stat)	0.0000
Time:	17:38:50	Distribution:	chi2(13)
Cov. Estimator:	robust		

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
prom	0.2648	0.0142	18.676	0.0000	0.2370	0.2926
product_1	-5.4225	0.0531	-102.16	0.0000	-5.5265	-5.3184
product_2	-4.4459	0.0763	-58.246	0.0000	-4.5955	-4.2963
product_3	-3.8220	0.1081	-35.352	0.0000	-4.0338	-3.6101
product_4	-5.9368	0.0468	-126.82	0.0000	-6.0286	-5.8451
product_5	-5.2923	0.0797	-66.391	0.0000	-5.4485	-5.1361
product_6	-4.5235	0.1256	-36.002	0.0000	-4.7698	-4.2773
product_7	-7.2170	0.0432	-167.18	0.0000	-7.3016	-7.1324
product_8	-6.9629	0.0566	-122.93	0.0000	-7.0739	-6.8518
product_9	-5.8604	0.0629	-93.166	0.0000	-5.9837	-5.7371
product_10	-6.9324	0.0337	-205.62	0.0000	-6.9985	-6.8664
product_11	-5.9682	0.0683	-87.364	0.0000	-6.1021	-5.8343
price	-0.5286	0.0153	-34.607	0.0000	-0.5586	-0.4987

Endogenous: price

Instruments: instrument

Robust Covariance (Heteroskedastic)

Debiased: False

2.4 Compute the mean own-price elasticities for all products (10 points)

6 points for correct calculation of elasticities

4 points for reasonable results

```
[11]: alpha = logit_modelb_iv.params.price
otc['elasticity'] = alpha * (otc['price']) * (1 - otc['share'])

# add this to our summary table so that we can easily compare
sum_elast = otc.groupby('product').agg(elasticity = ('elasticity', 'mean')).
    .reset_index()
summary = pd.merge(summary, sum_elast, how = 'outer')
summary.round(3)
```

```
[11]:   product  price  total_sales  wholesale_price  brand  size  \
0         1  3.427         51280           2.187  Tylenol   25
1         2  4.946         63604           3.681  Tylenol   50
2         3  7.028         41305           5.771  Tylenol  100
3         4  2.969         42032           2.027   Advil   25
4         5  5.146         27627           3.630   Advil   50
```

5	6	8.158	12703	6.104	Advil	100
6	7	2.674	14365	1.851	Bayer	25
7	8	3.619	12310	2.438	Bayer	50
8	9	3.971	28324	3.710	Bayer	100
9	10	1.935	33175	0.910	Store Brand	50
10	11	4.427	27208	1.914	Store Brand	100

	market_share	price_per_100	elasticity
0	0.145	13.708	-1.810
1	0.180	9.893	-2.612
2	0.117	7.028	-3.713
3	0.119	11.876	-1.568
4	0.078	10.291	-2.719
5	0.036	8.158	-4.312
6	0.041	10.697	-1.413
7	0.035	7.238	-1.913
8	0.080	3.971	-2.098
9	0.094	3.871	-1.023
10	0.077	4.427	-2.339

3 New Product Introduction

The chain of stores you have data from is considering introducing a 25 tablet size bottle.

3.1 Assume WTP for the new product will be equal to average WTP of the brand name 25 tablet products, minus the “store brand” effect you estimated in the first part above. Assume there are no promotions of the new product. What will be the expected demand for the new product at a price of \$2.00? (15 points)

6 points for correct calculation of WTP

6 points for correct calculation of new shares

3 points for reasonable results

```
[12]: # WTP = average product_fe - store_brand
beta_product = {}
for i in product_numbers:
    beta_product[i] = logit_modelb_iv.params.get(f"product_{i}", 1)
beta_fe = pd.Series(beta_product)

beta_storebrand = logit_modela_iv.params.store_brand

WTP = ((beta_product[1] + beta_product[4] + beta_product[7])/3) -
    beta_storebrand
```



```

[13]: # pull coefficients for use in calculating shares
beta_prom = logit_modelb_iv.params.prom
alpha = logit_modelb_iv.params.price

beta_product = {}
for i in product_numbers:
    beta_product[i] = logit_modelb_iv.params.get(f"product_{i}", 1)

beta_fe = pd.Series(beta_product)

[14]: # recalculate market shares with new good
# calculate logit share numerator for each good
numerator = {}
for i in product_numbers:
    otc_i = otc[otc['product'] == i]
    numerator[i] = np.exp((alpha * otc_i['price'].mean()) + (beta_prom *
    ↪otc_i['prom'].mean()) +
    (beta_storebrand * otc_i['store_brand'].mean()) +
    ↪beta_fe[i])
numerator_before_new_good = pd.Series(numerator)
numerator_after_new_good = pd.Series(numerator)

# add share of new good using WTP calculated above as product FE
numerator_after_new_good[12] = np.exp((alpha * 2) + (beta_prom * 0) +
    ↪(beta_storebrand * 1) + WTP)

# calculate denominators for before and after new good is introduced
denominator_before_new_good = 1 + numerator_before_new_good.sum()
denominator_after_new_good = 1 + numerator_after_new_good.sum()

[15]: # calculate total customers across all stores in all weeks
total_traffic = (otc.groupby(['store', 'week'])['count'].first()).sum()

# estimate sales of each good before and after new product introduction
new_market_shares = numerator_after_new_good / denominator_after_new_good
sales_new = new_market_shares * total_traffic
old_market_shares = numerator_before_new_good / denominator_before_new_good

# add in value of 0 for new product before it was introduced
old_market_shares[12] = 0
sales_old = old_market_shares * total_traffic

[16]: # put it all into a table
otc_new = pd.DataFrame({
    'Old Shares': old_market_shares.values,
    'Old Sales': sales_old.values.round(2),
    'New Shares': new_market_shares.values,

```

```

        'New Sales': sales_new.values.round(2)},
        index = new_market_shares.index
    )

    # rename the index to 'Products'
    otc_new.index.name = 'Products'
    otc_new

```

```

[16]:
      Old Shares  Old Sales  New Shares  New Sales
Products
1      0.000725   48283.29   0.000724   48249.15
2      0.000873   58139.10   0.000872   58098.00
3      0.000545   36317.32   0.000545   36291.64
4      0.000560   37299.86   0.000559   37273.48
5      0.000337   22447.71   0.000337   22431.84
6      0.000149    9907.89   0.000149    9900.88
7      0.000185   12334.48   0.000185   12325.76
8      0.000146    9695.65   0.000145    9688.80
9      0.000371   24705.69   0.000371   24688.23
10     0.000278   18531.23   0.000278   18518.13
11     0.000194   12915.60   0.000194   12906.47
12     0.000000     0.00    0.000707   47109.12

```

3.2 Break down the benefits and costs to the store of introducing this new format (assume wholesale price is \$1.00 and there are no fixed or other variable costs of the new product introduction). (10 points)

6 points for correct calculation of new market shares

4 points for reasonable results

```

[17]: # pull price and cost information and add in the new product
price = otc.groupby('product')['price'].mean()
cost = otc.groupby('product')['cost'].mean()
price[12] = 2
cost[12] = 1

# add to otc_new
otc_new['Price'] = price.values.round(2)
otc_new['Cost'] = cost.values.round(2)

```

```

[18]: # calculate profit before and after the new product
otc_new['Old Profit'] = ((otc_new['Price'] - otc_new['Cost']) * (otc_new['Old_
↵Sales'])).round(2)
otc_new['New Profit'] = ((otc_new['Price'] - otc_new['Cost']) * (otc_new['New_
↵Sales'])).round(2)

print(f'Profit before the new product: {otc_new['Old Profit'].sum().round(2)}')

```

```
print(f'Profit after the new product: {otc_new['New Profit'].sum().round(2)}')
print(f'Therefore, introducing the new product changes profit by_
↳{((otc_new['New Profit'].sum()) - (otc_new['Old Profit'].sum())).round(2)}')
```

Profit before the new product: 348673.5

Profit after the new product: 395536.09

Therefore, introducing the new product changes profit by 46862.59

4 Information Intervention

The chain of stores you have data from is considering a campaign to help educate customers that there is no efficacy difference between the brand name and “store brand” drugs.

4.1 Assume the campaign increases the WTP for the “store brands” by the full absolute value of the “store brand” effect you estimated in the first part. What is the net benefit and cost to the store? To consumers? (15 points)

6 points for correct calculation of WTP

6 points for correct calculation of new shares

3 points for reasonable results

```
[19]: # pull fe again (just in case)
beta_product = {}
for i in product_numbers:
    beta_product[i] = logit_modelb_iv.params.get(f"product_{i}", 1)
beta_fe = pd.Series(beta_product)

beta_storebrand = logit_modela_iv.params.store_brand

[20]: # calculate CS before change
# calculate logit share numerator for each good (should be the same)
numerator = {}
for i in product_numbers:
    otc_i = otc[otc['product'] == i]
    numerator[i] = np.exp((alpha * otc_i['price'].mean()) + (beta_prom *
↳otc_i['prom'].mean())
                                + beta_fe[i] + ((beta_storebrand) *
↳otc_i['store_brand'].mean()))

numerator_before = pd.Series(numerator)
denominator_before = 1 + numerator_before.sum()

# need to artificially increase the utility received by patients buying_
↳store-brand drugs
adjusted_denominator_before = denominator_before + (np.exp(np.
↳abs(beta_storebrand) * 2))
```

```
[21]: # recalculate WTP without the store-brand effect
beta_fe[10] = beta_fe[10] + np.abs(beta_storebrand)
beta_fe[11] = beta_fe[11] + np.abs(beta_storebrand)

[22]: # recalculate market shares with new WTP
# calculate logit share numerator for each good
numerator = {}
for i in product_numbers:
    otc_i = otc[otc['product'] == i]
    numerator[i] = np.exp((alpha * otc_i['price'].mean()) + (beta_prom *
    ↪otc_i['prom'].mean())
                        + beta_fe[i])

numerator_after = pd.Series(numerator)
denominator_after = 1 + numerator_after.sum()

[23]: # estimate sales of each good before and after information campaign
new_market_shares = numerator_after / denominator_after
sales_new = new_market_shares * total_traffic
old_market_shares = numerator_before / denominator_before
sales_old = old_market_shares * total_traffic

[24]: # put it all into a table
otc_info = pd.DataFrame({
    'Old Shares': old_market_shares.values,
    'Old Sales': sales_old.values.round(2),
    'New Shares': new_market_shares.values,
    'New Sales': sales_new.values.round(2)},
    index = new_market_shares.index
)

# Rename the index to 'Products'
otc_info.index.name = 'Products'
otc_info
```

```
[24]:
```

	Old Shares	Old Sales	New Shares	New Sales
Products				
1	0.000725	48283.29	0.000724	48267.20
2	0.000873	58139.10	0.000872	58119.73
3	0.000545	36317.32	0.000545	36305.22
4	0.000560	37299.86	0.000560	37287.43
5	0.000337	22447.71	0.000337	22440.23
6	0.000149	9907.89	0.000149	9904.59
7	0.000185	12334.48	0.000185	12330.37
8	0.000146	9695.65	0.000145	9692.42
9	0.000371	24705.69	0.000371	24697.46
10	0.000278	18531.23	0.000474	31609.69

11 0.000194 12915.60 0.000331 22030.82

```
[25]: # pull price and cost information and add in the new product
```

```
price = otc.groupby('product')['price'].mean()
```

```
cost = otc.groupby('product')['cost'].mean()
```

```
# add to otc_info
```

```
otc_info['Price'] = price.values.round(2)
```

```
otc_info['Cost'] = cost.values.round(2)
```

```
[26]: # calculate profit before and after the new product
```

```
otc_info['Old Profit'] = ((otc_info['Price'] - otc_info['Cost']) *  
    ↳(otc_info['Old Sales'])).round(2)
```

```
otc_info['New Profit'] = ((otc_info['Price'] - otc_info['Cost']) *  
    ↳(otc_info['New Sales'])).round(2)
```

```
print(f'Profit before the new product: {otc_info['Old Profit'].sum().round(2)}')
```

```
print(f'Profit after the new product: {otc_info['New Profit'].sum().round(2)}')
```

```
print(f'Therefore, introducing the new product changes profit by↳  
    ↳{((otc_info['New Profit'].sum()) - (otc_info['Old Profit'].sum())).  
    ↳round(2)}')
```

Profit before the new product: 348673.5

Profit after the new product: 385015.71

Therefore, introducing the new product changes profit by 36342.21

```
[27]: # CS = log(denominator)/alpha
```

```
CS_before = np.log(adjusted_denominator_before) / alpha
```

```
CS_after = np.log(denominator_after) / alpha
```

```
print(f'Consumer Surplus before the information campaign is {CS_before}')
```

```
print(f'Consumer Surplus after the information campaign is {CS_after}')
```

```
print(f'Consumer Surplus increases by {(CS_after - CS_before)}')
```

Consumer Surplus before the information campaign is -1.886442797688499

Consumer Surplus after the information campaign is -0.008898765290945184

Consumer Surplus increases by 1.8775440323975539